

ActInf Livestream #027.1 ~ “Active Inference: Applicability to Different Types of ...”

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all right welcome everybody to the active inference lab today it's active lab live stream number 27.1 on august 24th 2021 welcome to the active inference lab we are a participatory online lab that is communicating learning and practicing applied active inference you can find us at the links here on this slide this is recorded in an archived live stream so please provide us with feedback so that we can improve on our work all backgrounds and perspectives are welcome here and we'll follow good video etiquette for live streams so people can just raise their hand in jitsi if they want to speak at any time or write a question in the live chat this short link has a calendar of upcoming live streams and today we're here in 27.1 on the 24th and in august 31st we'll follow up with another discussion on this same paper and then we haven't yet decided what to read in the upcoming weeks so if you want to come on and discuss that would be cool today in active stream 27.1 our goal is to learn and discuss this very cool paper active inference applicability to different types of social organization explained through reference to industrial engineering and quality management by stephen fox and uh we're gonna go anywhere we want with the paper or questions and we're really appreciative to have steven here so that will be a cool discussion let's uh start with just introductions the non-authors can say what was interesting to them or what kind of brought them to this discussion and then it will be awesome to hear some context on the paper that we can take the discussion anywhere so i'm daniel i'm a postdoc in california and i'll pass it to dean yeah good morning my name is dean i'm i'm in calgary and uh thing i found really interesting is that this was the fastest turnaround time between the 0.0 which is a bit of a deconstruction and now this point one which is now going to be sort of put back together or reconstructed today and i just like the the idea that the sort of the feedback loop is really tight so i'm really kind of both curious and uh happy that that the feedback uh is such a quick turnaround and i'll pass it over to stephen sillett oh thank you dean yeah um yeah i'm just really intrigued i'm very excited to hear more questions about how active inference fits in at the larger social and applied scales and i know a

lot of us here it's kind of a bit of our thing is how does this get applied in different contexts so uh yeah thanks awesome agreed about the whiplash dot zero so stephen welcome please introduce yourself and then maybe give us some context hello everybody so my name's stephen fox i'm a principal scientist at the technical research center of finland and my background is i left school when i was 16 i was an apprentice and did that my craftsman master craftsman did manual work for 10 years and then when i was working full time in industry i got a phd in engineering design and subsequently i've got a doctor of science degree in production engineering and i came across um active inference and i thought it looked pretty straightforward and useful but then i started reading some scientific papers about it and it became more and more difficult to understand at least for me so i thought the only way to resolve this was if i wrote a paper about it and sent it off to peer review so three iterations and five thousand words of critique later it was published in the paper that you've read okay awesome i have a few thoughts on that and then anyone can raise their hand so one piece was that getting more difficult to understand active inference as you went along what dimension was that you know like what made you feel that way and is that different than other areas that you learned about um well the scientific papers about on the topic they are they're addressing particular audiences aren't they and so for example some papers are addressing computer scientists who are using established machine learning paradigms so the paper is directed to them and i guess the papers are sort of responses to criticisms that come from that audience so if a reader's not a member of that audience it um it's not clear what it's about so i think the core idea is uh straightforward enough but so it's sort of like this the like the relationship to between thermodynamic entropy and information entropy the the so this is beyond the act of inference literature the basic idea is simply enough but of course if you read a scientific paper about it the scientific paper about it is about some particular issue there when it's supplied in a very particular for a very particular purpose i think i understand that the relationship between thermodynamic entropy and information entropy but if i read any scientific paper about it i at least can become baffled because it's a very focused issue and of course in in scientific literature you need to go beyond the state of the art so i think that's the thing that and i asked a colleague of mine who uh to to look into the literature as well and he um [Music] he was baffled enough to decline in my invitation complete with a funding possibility to carry on looking into it so dean thanks for sharing that stephen so dean go for it and then anyone else with a question yeah stephen i was just curious so you you're kind of alluding to who's in the room that that you know who's the audience thing and i'm curious because you you sort of

understand yourself as being a bit of a craftsman and a bit of an artisan so do you think the paper has a ready audience or do you think your your paper can act as as an attractor state because we're talking about active inference um my own my own paper that you've been reading yeah um well well i think um well the first thing is now it's clear to me so uh and also it turned out i didn't know until it was published that the academic editor's professor fristen so at least it's clear to me what it is in what i'm doing and so i can apply it so for example i've put it into a proposal that's currently under review and hopefully get some funding and then if it's of use to others then that's good would be uh and i i've had one person who's uh read it very thoroughly and said it was he said it was a great help to him and he wished he this paper had existed three years ago because it would have saved him so much trouble of going on the treasure hunt through the literature so that's uh that gentleman is in in the world of business so he also had been on this treasure hunt through the literature finding a sentence here and a sentence there that he could sort of put together and think well do i understand this don't i understand it so yeah hopefully it's of help to others but it's not i mean i don't think there's a big there's not a big audience for this i don't think it's a niche it'll always be a niche um and it makes me think about how the development of this niche and the community dynamics within it have been different than maybe other fields like what you said again about getting more difficult or perceiving the the implications or the difficulty to be higher than when you initially jumped in first off is a is a meta learning characteristic you're updating your uncertainty or your scope or the policies that you're going to have to take as you explore the space and then there's um something about this field that impels a lot of people to learn theory and to explore applications whereas i don't know other related fields sometimes like you said they're either framed as a technical debate like people have shown this relationship under this condition here we show up for this other condition and then that's a bit too abstracted away from uh meaning for many people or it's just in some corner of the literature um or just sort of it remains at a qualitative sort of okay cool cybernetics like goal orientation and then it can veer off and maybe get disconnected from some of the formalisms and so you mentioned the core of active inference or that now you have a clearer sense of it so what did you initially or what is your more refined sense of what that is now illumination came to me when i read a paper i think it's an uncertainty and stress and it says um a professor fristen's one of the the authors and it says that free energy something like this the free energy is the information that somebody is lacking to make their internal model match reality and of course we know then we could be discussing forever and a day about what reality is i thought that was a so on the treasure hunt that's uh a

sparkling jewel of information because i thought okay so ah so it's an information gap it's an info what we're talking about is the information gap between the internal generative model the external state what the uh and this variational free energy because of course our internal state keeps changing and the external state of the world keeps changing whether we want it to or not as we try to uh and as actions haven't the actions we're gonna take we haven't taken them yet we seek to reduce expected free energy over a series of actions and so to me what what is that mean it means we we seek to reduce the information gap our internal generative model of the world and the world um which we are in that we're in and then active inferences well how do we do that we can the actions we can take we can either we can change what we pay attention to that's an action or we can take action in the world to change uh the sensory inputs so that's it and if so if now this reducing the information gap i'm as you'll notice in the paper i'm using this term unwanted surprise so surprise all is the difference what we are predicting and then the sensory input and the world itself these things can be for example in consumer culture survival may mean selecting the right garb and accessories so one can survive in a social group and prosper in there maintain one social capital so if we're in the if we're in the rich world survival can be at that level but if we're in the uh other parts of the world survival can be the difference between literally between energy output and energy input over the course of a week and you literally might not survive if you don't get that right so does that make sense very interesting description just to kind of trace out the um order that you brought up a few the variance of free energy in so you framed it as being free energy is the information gap and then the external states dynamics so that makes me think about like trading like if you knew what the shape of the stock or the crypto is going to look like you could trade perfectly and then for other situations it's also like if you knew perfectly what was going to happen you could act perfectly you could steer your ship so that given the exact wind distribution you are always like navigating at the fastest possible uh rate but that is kind of an intractable calculation to make due to for other reasons um the internal and external states changing and that's leading us to the variational free energy and then that is a statistical calculation conditioned on action which leads us to the expected free energy which is seeking to minimize free energy indirectly through expected free energy directly and then you pointed out that there was this capacity to change attention and to change sensory states through action so i think that was a very succinct uh run-through of a lot of the active inference things dinner stephen what did that make you think about well when i'm done could i just could i just read could i just read exactly what it says in the paper yeah uncertain to interest it says can be

regarded as the information a person is lacking and which uh he strokes she could use to make his her internal model as close as to reality and the authors of this paper uh then these are so there's professor frison but the other authors are being doing resurgence right so there's a bruce mcewen and carl fristen it's called stressing stress and uncertainty okay why it causes diseases and how it is mastered by the brain and it's published in progress in neurobiology got it thank cool so that was the paper that kind of gave you the insight into the of yes where where's the chat here i can send you the link in the chat um it's on the bottom left or you can email it afterwards and we'll add it to the youtube uh description so cool dean what were you um uh gonna say um on that no i'm just gonna ask i was just gonna ask uh stephen so when i was going through the paper until i got i don't know probably 80 in and started seeing the references that you made to the to the wayfinding aspect in the in the exploration piece prior to that there was a lot of there was a lot of attention given over to the precision piece around the free energy principle and what i was one of the things that i commented yet in the two-point 27-0 was if we're talking about active inference we're we're we're basically talking about two things at once we're talking about precision and we're also talking about provisioning and i think my my simple metaphor is are we talking about a hand that fits into a glove or are we talking about the provisioning piece which is how many different gloves have we provisioned in terms of the context that the hand might find itself cold weather water forging a piece of hot iron or whatever and and so what i was what i was curious about was in terms of the social organization in the industrial engineering piece how the how the provisioning and the precision coincide with one another in terms of what you are thinking now post hoc writing this paper can i can i share my screen somehow yes let me um unshare these slides so on the bottom left the middle button share your screen and then i'll make it um like full screen so you can make it look however you want on your site sure okay can you see him yeah hit the hide on jitsi where it says uh hide the sharing your screen and then um yeah i can make that full screen so thanks so you see a diagram yep figure one okay so why there's so much emphasis on what's going on in uh down here in make to stock and assemble towards is because what i was trying to do was get a an anchoring okay in here in where where there's um mass production and mass customization where there's a great deal of refinement of what is offered through many rapid iterations but then there's something that's fixed that okay there are there are a lot of activities like this in the world and the diagram about the quality process so this is in in a way this is as if we think about it every evolution like the toyota corolla is a highly evolved species compared to um if we make uh if we make a vehicle ourselves from scrap so i

don't know it's been in the news there but a teenage teenager in india has made a tuk-tuk out of scrap solar powered as well so then that's a it's not highly evolved it's the first prototype working prototype and then these things up here this uh top engineer to order the initial situated entropy is much higher and the barriers to information gain are much higher as the diagram you're looking at just changed yeah okay so this is from another so that that diagram i just showed a minute ago that's from another one of my papers in entropy and this is from another one so what's going on is is that um if so with uh diy dude do i do it with others presumption uh massive engineering but we it begins with a just an idea in our in our minds and we may begin at the level of uh i think this in the video did you show a carved piece so it might be you've got a a tree stump in your back garden or you might have a tree and start down there raw materials and you might shape it and then there can be this phenomena that well there's a dialoguing between the human doing the shaping and the piece of wood itself because the piece of wood has got its own properties you're not going to get a an aesthetically pleasing carving if you chisel against the grain right so so the grain that has been shaped through perhaps hundreds of years some of the trees there in california so there's a dialoguing between the the human doing the shaping and the the piece of wood itself shaping what the human does but then more from an industrial point of view uh if um anything that is bespoke so the term apparel is more inner the term bespoke is more appropriate in apparel than um engineer to order so bespoke is short for bespoken for and it meant be spoken for by an individual so the materials that were to be used would be spoken for by an individual and in today what happens with bespoke is that there'd be cloth so that's a formed material parts buttons then suppose it's show business or suppose that elton john in his most flamboyant days had an idea in his mind for a new uh costume it would be a preliminary idea there'd be iterations of sketching and uh doing some mock-ups maybe and looking at the fabrics and they'd do it and then maybe we get to quite an advanced stage of the fitting and he'd decide that he'd look at it and think i don't like this actually and completely change his mind and they do something else and then with engineer to order that could be for example uh the late steve jobs is yacht so he might have an idea in his mind about uh what it might be what it might look like and then the the ship builders they're going to try and get as far up the bill of materials as they can so they're going to try to use the um the cabin modules that they've used on other ships or at the very least to use the uh the 3d digital models that they've got what's in the paper it's really around this um it's it's down here because then that's the so rather so to start with something that's quite a concrete and highly transferable before then as i do later in

the manuscript in the paper getting into the project processes the way finding and so on does that answer the question can i just do a quick follow-up just so it's clear in my mind um so what i'm seeing here what you put up on the screen is and i don't i don't i want to be corrected if i if i haven't got this right is i see that i see ways of laying out a a fabrication series and i guess what i was wondering is more on a philosophical level as you're transferring from idea to fabrication how do we use active inference there how do we turn the idea and translate it into this fabrication exercise is there is it a is it is it a planning effort is it a is it a getting our ideas plural provisioned is it about propositioning the the garment maker before they they go on to to do the construction is it you know i i i what i'm trying to do is get in my head clear what your focus was on because i could i can tell now already 40 minutes into our conversation today that there was much more you could have written about but i'm i'm sure basically what you wanted to do was was get one one thing clear in terms of this paper but that for for people like me that opens up a whole pandora's box so i just want to be in my mind certain that the focus here is about the fabrication piece and not the stuff that comes before the fabrication piece of my am i leaning in the right direction seems like he silently left actually oh no sorry oh no i was on mute oh continue then yes yeah so in the in the paper itself um i think i'm using the term it's mass production and project production is it yeah so yes other the focus on the paper of the paper is on this mass production to just to try to what's the word i'm looking for i can't think of anything better but just just just to sort that out just say okay that's what happens in mass production and here are these diagrams and these explanations here and there's the the um statistical process control charts and things like that that's what goes on there that's settled now we move on to these project the project production and why i'm focusing on physical production is because it's then it's tied into um action it's the there's a clear link between action and energy and of course energies in this is a finite supply of it uh for individuals and organizations the mass production stuff is to provide the the platform to go into more challenging areas am i answering your question dean tangential rambling no no no no like it i guess it depends on how much you desire decide to zoom out um and so i i just want to be clear in my head what the focus is on because there is absolutely active inference in fabrication and manufacturing i have no no quarrel with that whatsoever i just want to be able to make sure that i don't overextend what i think is the act of potential because as ever said when i was reading where it got exciting for me was that whole exploration beyond the precision and on to the provisioning or provisionings part so thank you yes you you cleared that up for me awesome how about stephen with ray's hand and then anyone else yeah thank

you um i i'm i'm looking at this here and i see some interesting um ways that you're engaging free energy in relation to active inference because often we we might see mathematical models and then philosophical arguments and i sense that this mid-round which may be where a lot of the applied work happens at scale is is heuristic based or principles based to sort of get a sense of how to navigate and may not be directly modelable because there's so many dimensions so that's kind of interesting so i wonder you know where where people are talking there about situated entropy or the the the it's it's a potential so i'm wondering how you how you think about the heuristic and how you think how much someone could model this type of thing or how much it's about modelling and if so what type of modeling might be useful in organizational industry contexts well i think at the end of the the paper i i raised this question i think about there's uh there's maths and then there's modeling and then there's uh simple rules and so in engineering design and not just engineering design uh simple rules can be extremely so designed for manufacturing assembly the application of that has led to 90 reductions in the assembly times of many products you could you could have just kept modeling the the former process as people did for for many years with ever more sophisticated mathematical models and simulation tools but it still still wasn't very good or you could just apply some simple rules reduce assembly time by 90 percent well then it's a lot better isn't it so i'm i sat um in one of the uh the meetings that um professor fristen and maxwell ramstad i think we're going through professors fristen's book the monograph the name of which escapes me now a free energy principle for a particular physics professor fristen and thomas parr were discussion with is it michael bail who's uh he works for a japanese ai firm uh martin deal yes money bill yeah and um at least one colleague of martins and they were discussing amongst themselves i don't think as far as i could tell they weren't able to get an intercepted understanding between themselves even and martin's published a paper in entropy where he's contesting some of the maths my view is that of course things develop and they have to start somewhere and no doubt that will lead to good things but it would the more people that can understand something the more likely it opportunities for its development and flaws in its current um framing can be identified and progressed i'm and i've sent him my email this morning this quote from einstein didn't know from from geometry and experience that what do you have it there now um look at that quote from al einstein it was in my email um i can look it up in just one second but right now just on the screen was the um guest stream with martin and so totally agreed really fascinating there um okay answer the question i'm i see of course there's there's a place for maths there's a place for computer simulation but if so if we go for so it may be that some are

engaged in positivistic research and they're they're seeking to establish a correspondence truth and others you know anti-positivists interpreters they are engaged in looking for constructivist truth and what i'm doing is i suppose it's critical realist pragmatic truth so if it works it's true what i'm doing is well it's probably a bit rough but maybe it's good enough to get something done yep very interesting um one question i had was you talked about that kind of co-evolution with the iterations whether it's the the craftsperson working with a raw material with its imperfections or its characteristics and then all the way up to this more social form which is like uh elton john and the costume designers in in that feedback so whether it's just hands on object like drawing or whether it's it's more extended to include all these other aspects hold space for that versioning and co-evolving process and for the material to allow it to reveal its characteristics while also having effective supply chains you know that can make cpus and that deliver high precision materials because it just sounded like you had experience on on both sides with the industrial engineering as well as the craft and artists inside yes so um with uh automated engineering tools yeah as you i'm sure you know there are these communities um of makers and uh where people are sharing is instructable still going as a online community of giving instructions for projects and things so so one one big challenge is that it's skill so manual skill if work like carving it depends upon um quite a lot of manual skill to get a reasonable looking carving that somebody might be happy with but if you're going to make something if you're going to 3d print something the the manual skill required isn't so so high and that can be communicated so some of the uh work that i've been doing is in this virtual social physical convergence through virtual technology you can share socially skill knowledge in doing physical work if also um do you know the open ecology movement where there are engineering designs uh freely available on the internet so through virtual social physical there's great potential to enabling more people produce what they consume to become prosumers and as gandhi i paraphrase gandhi now he said prosperity can't come from mass production it can only come by from production by the masses interesting answer reminds me a lot of bucky fuller trying to pivot the supply chain from weaponry to livingry mass producing the aluminum sheets for housing but that was kind of only potentially a transient step in this supply chain in production and then eventually knowledge reimagining will we all have a silicon foundry in our house probably not could we all have a compost bin or could we all have a sewing kit that's something that's a totally different type of artifact so it can be mapped differently dean and then anyone else or any questions in live chat well i don't want to go off into into the woods here but i was curious uh stephen if if you've ever read any of this

stuff matthew crawford in particular shopkraft title and then the other one the world beyond your head and the reason why i'm asking is because in that in that second book which is one of my favorites he he talks about all of the distractions in the world that cause us to not be able to be focused and precise and do the the crafting never mind being able to do the the uh indus you know projection at an industrial scale and so i just i was just curious if if maybe there was um maybe that author was maybe whispering in your ear when you were putting this together because if you whether you're whether you're carving something out of a piece of wood and having a dialogue with it which i completely agree with or you're just making sure that there's no bugs in the in the command software for the for the making of the copies it's still an ability to be able to use things like active he wrote he wrote the beyond your head book in 2014 so i don't think he was looking at active inference but i think how that is employed can sometimes be perceived as just the right amount of being able to give attention over to something and still remain open to possibility so i was just wondering if that was something that may have been uh adding some color to what you were doing no i had not heard of that before but i shall i shall look into it with interest yeah it's fantastic stuff yeah thanks for the comment dean stephen yeah this um i hope this does again doesn't go in the weeds either but but i think this is like say this this touches on so many things is when you talk about the generative model and you've got in body cognition sort of fitting into that um generative models often be thinking that it's an action orientated model so it's an act it's a model of how to act and i'm curious how far for instance if you do welding you mentioned welding and stuff like that with this skilled performance idea that can be the generative model of how to weld but at what point or this sort of maybe ties into something i've been exploring at what point do you think that the instructions to weld are in the niche and being used by someone's generative model of which they only have a certain sequence available you know um now welding it may be that someone just they get a skillful practice and go the whole way you know so it's all in there but i'm just wondering once you get to sort of this management level at what point it's artifact as much as it is a generative model internally well i think uh with craft skills it's being able to do it it's just at some point or other after as well as an apprentice i think most people are apprentices have had this feeling that you're struggling and then you wake up one morning and then suddenly you've you've got a feel for it you can do it exactly how that is i don't think anybody really knows because somehow it's going into the the fascia system that's the thing the feel for it is embodied in the fascia system and most to my reading as a craftsperson most literature about embodied cognition is just paying lip service to it being about embodied

cognition and it's not really addressing the nature of embodied cognition about how much that depends upon the fascia system and body memory in this 4e model about extended cognition you know put put your memories in your mobile phone type of thing but there's no way that you can put your your manual skill uh embodied cognition muscle memory in a mobile phone you can have many videos of people doing manual work from all kinds of different angles so that can be informative but it's not uh as as you can have many videos of people riding bicycles that's not the same as having your body memory of how to get on a bike and not fall off while you're pedaling it um yeah stephen go further than i ever thought yeah i think this is a good i i agree i mean there's so there's some transition that happens when it goes from presumably some instructional process which is being decoded in a in a kind of a cognitive way to something more embodied like you say and it becomes a practice but then there's also i suppose and i don't know if but when you start coordinating for instance you've got a whole group of welders working on the side of a oil tanker so there's there's kind of then it might start to move again there's a scale beyond the individual welder you know and at what point does it then get beyond what i've been thinking of as like an experiential ecosystemic kind of knowing something which has to be coordinated through some artifact like a plan or something um let's just go to blue first go ahead because your thought is probably my thought oh no you should go for it so i have a question that was kind of related to steven's question um in terms the niche modification um as not as it relates to craftsmanship per se but to industrial engineering and and in the this paper that um was the topic of our discussion you said that um you know the the processes need to be carried out right the first time in like 99 of the cases so i remember this and and to what extent is that first time right i mean i i just remember as a business owner there's a lot of sop development right so you develop sops and kpis so that everybody knows exactly what their job is exactly what they're supposed to do and exactly the procedure for executing the job so that theoretically anyone can come in and be like subsumed under the markov blanket for the business and the niche has been modified maybe in such a way to make that happen correctly 99 of the time the first time so do you think that that process development in in the industrial engineering application is niche modification or perhaps something else um yeah so this the 99 times out of a hundred that's so that's six sigma which is under the umbrella of total quality management that is where uh so if you if you've made so this is all by the way are you still seeing my screen uh no uh maybe could you reshare it okay yeah hang on so let me just if i go back to um so if i go back to that point i was making about some embodied cognition yep looks good this this can you see that certain amount of

predictive processing yep what i'm doing in this paper is i'm uh refer i'm making reference to the fascia system and body memory and relating you know well predictive global neural weight workspace that's um as far as i've got with that now this 99 now can you see the treadmill yeah okay there it is this this 99 uh at a time right this time this has come with the treadmill of production and consumption so if if an organization invests a billion dollars in building a factory and uh and then the tooling it's got to mass produce and it's got to get people to to buy what it's selling and they can't be disgruntled customers because now there's global competition that 99 it can be measured to microns with car body panels and things like that but if if you're in uh the leather the leather cluster in italy and you're making leather bags as has been going on for hundreds of years the precision is not so desirable it's good if every one of the leather bags is made that is made is a little bit different to each other because that signals that it's handmade so the 99 right first time he's all wrapped up in this um treadmill of production and consumption with varying levels of precision depending on what the niche is so if the niche is you're a global car maker the door better close with the obligatory clunk and it better be a water tight airtight fit when you're driving at maximum speed otherwise customers will be lost in different niches the level of vision that's required depends upon what customers are expecting and what the competition will tolerate did that did i answer the question at all thanks for the interesting answer i'll all give a thought and then we'll go to scott so first um it's almost like what about a semantic precision like a writer who is outputting um precision in terms at multiple levels like a typographical precision and also um like they don't just want to release you know their hand hitting the keyboard that's not what their writing product is so it's not just the mechanical precision that we can talk about and then that idea of waking up to skill and the way you connected it to embodiment is really interesting maybe people wouldn't expect that from an industrial designer but then when we think about craft in the continuum it makes more sense and then also that waking up aspect of it it's like during sleep when we're simplifying and pruning neural connections the more neural applications of free energy principle in active inference could help us see that as like a process of model simplification and unlearning the details and that is sort of where skill and mastery uh comes in and that could be cognitive unlearnings and bodily updatings in terms of the morphogenesis of the tendons or the sensory receptors or all kinds of other changes and so it's just cool that we can like really telescope in on just wanting to apply it to as you did in the paper really partitioning out the biological systems partitioning out the flesh and bone but we know we can go there but there's there's so much that you can say about just the engineering side that it was

just like it makes it very interpretable yet also clearly comprehensive which is what you pointed out was kind of lacking in the literature when you were learning so scott um go for the question i don't know if you watched the dot zero but we we we uh we we had a scott david alert with the markov blankets of contracts and but yeah anyway just go for it yeah thank you sorry i was delayed i was on another call a couple of things that thank you a couple of interesting comments that um sparked some questions ahead so the talking about the fascia and the and your statement about the mobile phone can't record the fashion obviously that's true but it sparked three things to me i wanted to get your thoughts on one is what in the notion of the fascia and the embodiment is it possible to use active inference framings or is it fruitful in the opposition of various fascia elements so when you're learning diving or frisbee playing one muscle is pushing against a bone or whatever and so the fascia has multiple components as well and so are there learnings that happen among fascia elements themselves because they're building actin myosin you know bulk or whatever i guess those are the proteins but in any event they're building bulk that's one is the opposition among fascia elements the other one is it reminded me a little bit on the mobile phone thing sandy pentland at mit talks about i think he did some work on predicting if people are going to get alzheimer's by they have their mobile phone in their pocket and they start walking around their apartment differently and the mobile phone is able to track that there's a difference an anomaly and so i thought that was kind of interesting that although the mobile phone can't track your craft your but it can note anomalies based on picking up fascia histories and then the third thing it made me think a little bit about um robotic prosthesis and haptic sensors and so there again you have this feedback mechanisms that are bounded obviously again they're not storing craft but it seems like i just wondered if any of those sparked any um thoughts relative that question of the i guess not just the mobile phone but that sensory apparatus and the information networks being responsive with elements of fascia in different direct and indirect ways and maybe an active inference framing a hybrid between the technology embodying a certain expectation let's say and the fascia embodying a certain expectation and that kind of harmonic coupling if that had any any thoughts on that thank you um i think one thing that i'd consider in all these things is what's the i'm going to mention the r word now reward uh what's the uh what would be the effort to reward ratio of doing that so if if um like you said about learning to dive so i'd say that maybe there would be some uh rules sort of like design rules for learning to dive that maybe could be formulated that might be useful for people to learning to dive but then going into very sophisticated modeling when individuals are so different that might require a huge amount of

effort and the reward may be less so i don't think you're alone when i was saying before scott that uh that in industrial engineering so up until 1980 people use the mathematical models and the simulation tools available to to to model uh industrial processes designed for manufacturing assembly was introduced and assembly times in many products was reduced by 90 percent with simple rules and so all the amount of there's there's a place for modeling but what i'm more interested in phase focusing on are there simple methodologies tables and diagrams can provide insights as designed for manufacturing assembly did that can lead to very substantial improvements so that's that's what i think but but then like this gate tracking you were talking about about people are they are they getting out on is that one thing that i think's very attractive about active inferences it can provide a starting point where humans and artificial intelligence can be starting from the same place that okay what what's going on here is we're trying to reduce this information gap the internal generative model of the human being the external state of the world that they're moving around in both the human being trying to do the under and the artificial intelligence is trying to do that and how is that done it's done through we learn either changing what we pay attention to and or changing the action we take so then there can be a common starting point and maybe this can make the human being feel much more comfortable around the artificial intelligence so i've got i'm working on a project at the moment that's this third wave of ai where there's some top down rules bottom up sensory inputs combining i see active inferences being well worth exploring as a starting point human ai interaction where both the human and the ai have got these well this this generative model that's the humans it's it's it's understandable to the awesome points about where active inference could come into play and also almost from a designer's perspective if they have a systems diagram uh or a digital twin of some type that includes active inference agents interacting first because they're all of a similar type of model they'll be increased interoperability and ease of computing these kinds of systems the system designer or perturber can simulating with counterfactuals or with any other technique can uh formalize what affordances they do and don't have so it's like if someone says the only way i'm going to get this marketing um you know to succeed is if i update their generative models so that they think that you know this isn't a ponzi scheme it's like if that can be formalized then we could have benevolent shapings of the niche and kinds of interjections like we're going to inject this much money with this much of an equation because we're doing kl optimal control on this parameter conditioned on these other parameters you know how do people feel about that versus we're gonna do an ad hoc strategy based upon conversations that you weren't part and then that just

while you were saying about the generalized principles again really interesting because there was a lot of detail in the paper and a lot of really insightful comments but it wasn't um the same formalism-driven approach that we've seen in some other works directly of course cited a lot of the primary literature there so it's just cool how you pulled back to the level of the generalized principles and then we can dip down into more special mathematical framings but this last part of table 1. which were also all sections of the paper or like essential kind of principles at least you and the reviewers agreed were important enough to define clearly yeah i found that that quote now so this is from albert einstein geometry and experience addressed to the prussian academy of sciences in berlin 27th of january 1921 as far as the law of mathematics refer to reality they are not certain and as far as they are certain they do not refer to reality nice that reminds me also the way that you again describe free energy as that it's almost like we can have a description or an estimation of of a gap a negative space and so it's like well what is it what's what's in the gap it's like well if i knew or if we had any more precision then we'd tell you but all we know is how large the deviation is uh so it's sort of like a way of uh one way around it is like using a positive definition of something that's lacking and then that's closely related to this idea of tractable minimization that is better suited to be calculated than an intractable maximization like maximization of reward or expected utility which is a totally different set of constraints and outcomes than a variational free energy gradient descent so scott and then anyone else i wanted to you know that got me thinking are they just the uh i guess the um library of opportunities for a system or i'm not sure how to characterize it but um one of the things i've been fussing with is the chris christopher alexander um for folks who haven't been in that space with i just learned that a few months ago about him but he's an architect who collected the idea um that he had had was the collector uh he was in in architecture and is collecting up different practices in architecture and different patterns and he came up with a pattern language the idea was if you want to create a portico and a place to negotiate private and public space for instance you had certain patterns that happen in different kinds of architecture and so if you wanted to do architecture you could look at a bunch of patterns and and stare at them for a while and see which one look best and it's interesting with these questions of filling gaps you know the easiest place to look is what other similarly situated parties have done in similar situations so if you have that crosstalk the peer-to-peer conversation um then you can discover without having to go through all the pain and the cost of discovering yourself and that's the notion of guilds right i heard reference earlier to apprenticeship learned by the masters not just reinventing the wheel kind of idea

okay and it's interesting when we talk about gap filling because in more static systems culturally static systems let's say historically quiescent systems periods then you can learn a lot from the guilds but in periods of high dynamic change environmental change the guilds are going to maybe under serve you because there may be so much change that the old practices just don't hold up the externalities are such they swamp things and so i wonder in that gap if there's an element of discovery both of the available what what are the ways to discover what's available and then discover the things that are not available a because they're in a different silo so they haven't been available but maybe because they haven't yet been tried in any silo right so they're just not they're emerging type of notions and maybe that's a bad question i don't know it just it felt like that gap there's there's an easy low-hanging fruit of practices chris alexander kind of stuff is relatively low hanging and then it seems like there's other mechanisms here of discovery and maybe that's randomness and maybe that has to do with evolution but anyway if that's commentable i'd like some comments on that yeah awesome question scott um are you are you still seeing my screen yes i'll make it big all right okay so now skills leapfrog skills yes there it is so um the thing is about manual skills as scott was just saying the guilds and everything so in the work that i do development projects that point out that industrial thinking is 300 years old it came from the english midlands 300 years ago and and taking that as the uh there's the basis of um doing development work may not be a good idea because it's 300 years old and the world's change and so it is with skills it's good to be able to carve wooden pieces it can be very rewarding and maybe it can bring a few people some economic benefits but that can't uh address the poverty of billions of craft guilds as i say to people well when they say ministers when they say well we need to build training centers and give people apprenticeships and i point out well that's medieval thinking isn't good that's not very good so here there's explanation of leapfrog skills i'd take you down to some diagrams here oops just a simple example bear with me this is taking longer than were you were you what was your active inference uh stage or awareness at this time in 2016 or when you're writing it leading up to here i didn't well no but the thing is it looks it depends how you look at it doesn't it because i so if you're doing manual work then you're doing active inference it what what oh what are we doing this uh right so information gap how am i going to do this because i need to do this because if i don't do this i'm not going to get paid if i don't get paid i can't buy food i can't pay the rent so how am i going to do that that's the and then so i'm going to pay attention my priors what have i done before i'm going to pay attention to this the sensory inputs from the external state and notice that oh that the not all the materials i need are

here so i need to improvise adaptive expertise i need to not apply what i've done before where it's not gonna help and i'm gonna have to think of a bit of something to do it and uh so in my opinion i active inference is certainly what's going on in manual work and then you you give it a try it doesn't uh all right it works to some extent or not to another extent so then look back in the generative model hang on what have i done i've done something like this before look around oh look i see in the distance there's some bit of kit i can use over there i'll use that and so on um but to go back to this craft uh thing that can you see that figure five that yep the kind of stuff you put around the door so doing these joints the difficult form to fix very often if you stare at such joints closely you you'll see they'll have been filled up with glue and dust or a bit of more paint being stuck in there but then if you do this instead it actually looks better it doesn't no need for the craft gill guild it's just a kit it's just um these molds it's just a kit so you've got the and these uh blocks here in the corner it makes it look more fancy and you can just assemble it so if i go back to this one do you see now a diagram that says movable factories yep movable factories on the job training product kits in the middle uh engineering design of capable processes for mistake proof production though this is as we know ikea spread around the world much more successful than any furniture company its competitors because they do engineering design of capable processes for mistake proof production and this can be applied to many things water towers solar arrays really so this in uh what scott was saying about the craft guilds that yes craft guilds although i'm a master craftsman myself it's medieval thinking and it's um it's just not scalable and it's very very wasteful hugely wasteful and really craft skills good for coping with chaos but chaos is hugely wasteful awash with entropy that can't be resolved so in in when i'm doing development projects this is what i'm advocating what's in that diagram uh dean with a raised hand and then anyone else who raises their hand i'm just throwing down a couple of things here so it's kind of it's in this is a nice fit into the glove of what scott was asking and then what i saw in terms of the title of that paper so my question is this if we um using the same kind of triangular shape as the five why's if we start at the top with learning and that learning then breaking down into one aspect of which is updating and then that breaking down into one act of which is gap respecting that then breaking down into assessing what the step function is and then that breaking down finally into some sort of action taken which is the filling to follow or the gap closing is that medieval thinking or is that an invention gradient descent i know that's a lot of stuff but if if you want to take the best of what happened 300 years ago and not get locked into starting with i'm going to close a gap because i perceive a gap present is there a way that we can take the best the past

meld it with the current and come up with an inventive gradient descent which which doesn't sort of hold us to either the craft or the mass i believe we can we'll be good friends yes and in um in that let's see if we can get to this there's yep uncertainty and stress nice questions what you said about the scalability of the mentorship in the guild model versus alternative models was really interesting because there's the uh sort of academia versus non-academia divide so with that sort of student through phd and that traditional mentorship system and how that's evolving then there's the traditional um trajectory the apprentice and the master and the way that that is also changing and the way that academia and research reconnecting so there's just so many cool threads about how everybody will find guidance so are you are you seeing my screen yep mass imagineering mass customization mass production yep so i'm i'm looking at in this one i'm looking at it from the point of view of consumer culture and saying how these complement each other it's in their deen about how these can all be complementing each other i'm definitely going to have a read of remember to maintain firm downward pressure on the soles of your feet while you do and pump to the tempo of staying alive no no no you've got to keep your feet on the ground otherwise it'll blow your socks off yeah i want to get back to um that was you struck me with that statement before the craft skills are very wasteful and they're useful with chaotic situations but chaos is really wasteful that's really that struck me and i wanted to ask you i thought well that's interesting because inefficiency is resilient and distributed makes me think about distributed agriculture distributed manufacturing a lot of the resilience discussions that are going on now i work with a group called all fed that's trying to figure out how to feed everyone in the world and things like that and then also the notions of monoculture and the production of cheap industrial calories but then there's no variety and again getting back to this the notion in active inference i keep coming back to carnot's characterization in the second law of thermodynamics and you need the hot and cold differentials to perform work with a heat engine and i feel like you need the differentials and information to have active inference bayesian occurrence that inefficiency feels like it's laden with value because of the differentials that are in there the chaos and the it feels like there's lots of value to be derived from chaos and inefficiency and maybe that's part of what it was well i guess it was being done in the industrial revolution notions as you took craft skills and reduced the inefficiencies and derived value and it feels like as we have more chaos in the information sphere are we how might we release those that value but but still recognizing that maybe it's a there's a font of inefficiency that keeps coming at us because interaction volumes are going up so we don't have to worry about inexhaustible supply of inefficiency that awaits

us but if we mine the inefficiency and bring it into the system where it's known then um it seems like it creates value i wonder is any of that commentable i hope that made some sense yes yeah yeah um but before i go on to that also dean there's this paper domesticating artificial intelligence can you is that on the screen yeah okay so expanding human self expression through applications of artificial intelligence in presumption so that's another um another one about this combining different different uh paradigms of production and presumption and yes so what i was saying about um inefficiency perhaps i should explain better what i mean redundant variation so redundant variation doesn't add it consumes energy and it doesn't add any value so that that is the the thing that with this leapfrog skills i'm pointing that out that well it serves no useful purpose to have redundant variation variation can have value but not not redundant variation so that's what i mean and also it depends on what people's circumstances are so if people are living on the edge as many people are there's just no they don't have any spare energy and redundant variation can uh well we know do you know this term the prokaryote that people in in the west that uh that people don't have they have zero hours contracts and and things like that and they're in the precarious but the real precare in the world are the people who were in developing countries you know they're they're pretty they're they're staying alive it is literally precarious that they they very don't getting water's difficult and when they get water it's dirty so they've got a choice they can drink it to keep living but if they drink it they're gonna get ill and maybe the illness is gonna kill them there are um different scenarios and in in development it's uh redundant variation is just trapping can keep people trapped in desperate poverty could you just comment a little bit on as a follow-up on the expand on that redundant variation what that looks like in some examples and also just wanted to mention in one of the groups i'm working with all fed we're thinking of developing a how many days to starvation app for people and like the countdown clock and the bulletin of atomic scientists but how many days a person has in their food availability including what's in the refrigerator what's in their food desert etc but it's days to starvation because of that awareness that precariousness but anyway but if you could talk a little bit more about redundant variations i'm not quite sure the concept of what that looks like can you see this diagram again on my screen this yes so in that um in the difficult to form fit and fix on the left hand side the shape if there's a little bit of a twist and um a little bit of poor machining in the timber sections there it can make forming that joint really difficult and then um those so those variations they're just they're just redundant they don't they don't add any value they lead to uh because there is this this the duality of entropy there's a little bit of physical disorder because

maybe the wood's a bit twisted or the machining was a bit off so there's the information uncertainty how am i going to make that joint suppose suppose there's six ways of forming that joint with equal probability just like a fair die and the uh the information entropy is 2.58 bits and that 2.58 bits it's just it's just not adding any value at all um that gap of 2.58 bit but with this alternative on the right hand side those uh the square pieces in the corners they can have a rebate formed in them in their in their thickness so there's quite a substantial overlap there's maybe uh nine millimeters or three eighths of an inch overlap so if if the joint isn't perfect it just covers it up and it doesn't matter so that's a pr a practical example and then if we scale that up to all the the many many things that can be so why uh design management factor and assembly reduce products assembly times by 90 is because it just eliminated or does that help you know what what that reminds me of is almost like we talk about aberrant precision we've talked about that on the neuro psychiatry side delusional beliefs um failure modes of decision making algorithms it's okay to not be latched on to the perfect strategy as long as there's this healthy uncertainty about strategic decision making because it's that uncertainty that's getting kind of coasted on that should the system persist ends up finding an appropriate or an adequate strategy but aberrant precision when there's like high precision um confidence but then also it's off target with its mean estimation things can go awry and so it's like that door frame on the left the mass produced as you've pointed out it's kind of like a false sense of precision because if you look at it it looks like the more precise one it looks like oh it's sharper or it was machined such and such angle at exactly this angle but then that one degree tolerance which sounds like really accurate on the machining wheel now that plays out on the door frame and now it's actually less precise in practice and also like you pointed out the example with the cupboard or the bookshelf being attached to the wall like you can design for internal precision really well but then it's hard to take a step back and have latitude for like the wiggle room of real life but that's actually where the success or the failure of the product in use is going to be so that's very interesting topics that you brought up you know just just a quick follow-up my happiest one of my happiest days in my life is when i discovered that you can mix sawdust with epoxy and then do gap filling and i actually completed projects a lot faster as a result one other comment on that is contracts mass contracts we had a lot of negotiations i used to work with banks and software companies and things doing contracts and it was funny because in negotiations a mass contract has that you get rid of redundant variation right you just get there's no reason to have a different contract for all these things but it constrains the people who have less power in the negotiation but it's funny because people say to you when

you're in the negotiation with someone who has a mass contract they'll say well you can't change it this is the way we we always have done it and that right they're saying this is just it and then the answer back to that is well let me show you how we're going to do it for the next 20 years you know because it's a world changes around these things that have insufficient variation in them and so it's interesting when what seems like a system that has eliminated redundant variation becomes anachronistic are you seeing my screen though are you seeing three circles yes okay so scott you were talking about uh distributed so what working on is this it's this highly distributed presumption highly distributed and highly efficient compared to distributed subsistence which is distributed but not efficient and centralized which is efficient but not distributed so that's the thing and here is the dark picture of what happens with centralization that looks like the fovea and retina the focus all the attention is on the uh the heart of the empire and then it's all dark so nice images and also that distributed model is very cool so steven with a raised hand and then anyone else who wants to raise their hand or a question in the chat i'd like to bring in a bit more on that aesthetics and embodiment i put a couple of notes actually in the in the in the youtube chat as well just uh with the link i i don't know if you've heard of the william morris and philip webb red house in bexley heath ken um it talks a lot about the crafts because they actually um they had this whole thing in the 1800s where they were wanting to redo the crafts in the in the house and actually had little holes in the ceiling so they could redesign the ceilings in a craft way for different events so there's this push against industrialization so i'm wondering when you mention about the better aesthetics by putting the corners i mean in some ways you you you gain what could be seen at one level of aesthetic but there again with a craft and seeing some of the variation the the the reach and i wonder how you feel about this the reach of the embodied of the person making it is is lessened so the ability to i wonder what your thoughts are about the ability to infer from the other direction what the process of creation was and how that aesthetic creates more soul how that is in a way what the soul of objects are let me try to sum up what my understanding is of what you what you were saying so my understanding of what you're saying correct me if i'm wrong is that if something is crafted then it has more inherent value at least to the person doing the who crafted it so if somebody carves a carves a sculpture that has more inherent value to them than if they downloaded a model of a sculpture and 3d printed it is that yeah there's and and on top of that the ability for their embodiment of creation their process of creation or their active inference or an active inference process of creation extends further out into that distributed network because someone who's further removed from that person could

see that sculpture and they could get closer to being them than being closer to being a cdc routing machine operator well i don't know because if i see it if i see a sculpture i can appreciate it hopefully but i'm far from being able to do it myself and uh so let me give you a more mundane example in many parts of the world walls used to have walls and ceilings used to have wet plaster applied to them to make them smooth plastering a ceiling getting the plaster on the ceiling the wet plaster on the ceiling and it's smooth it's a very difficult thing to do and uh so people will do apprenticeships as plasterers and so on so i can see that i can see oh yeah that looks and maybe i could try to do it myself as well and all the plaster would end up on the floor and on me and then i'd realize that oh this is really difficult it doesn't get me any closer to being a plasterer with the skill to to get the plaster on the ceiling and if all dwellings depend upon the construction of all dwellings depends upon such processes billions of people are going to remain main in bad dwellings or no dwellings so that's the so if i understood you correctly i don't think it makes me but if i see somebody putting up um some boards on a ceiling that i've designed to give a a functionally um adequate and aesthetically adequate ceiling and i see them do that then i can do it cool thanks just one just one just one follow-up just very very quickly i think i suppose i could also i'm not necessarily disagreeing with what you're saying at all i think but i would also in terms of this distributed um like if you go into a medieval house and you see a more rough and ready plastering you may not know how to do it but you could imagine the process of someone in bodies that did it even if you couldn't do it yourself and so i was wondering about that those kind of those removals as things get industrialized which may be you know maybe that's why sometimes we add texture back in and sometimes more expensive stuff that we buy in the shops actually has texture put back in that can be actually expensive to produce it reminds me of graffiti like when you see graffiti you can imagine an affordance that would lead to you having done that art versus like a you know an apple phone it's delivered like in the conch shell with smooth and you know it's sort of the extreme case uh where at the very least it's more cognitive to understand how it was built and it definitely makes sense and things that are well worn they uh it doesn't make them like better or more valuable just makes them more able to play a part of a story in someone's life like putting up a kid's drawing it becomes about the person and latching on to a broader network of connections versus the very parsimonious and ruthless efficiency of an industrial design process that kind of seeks to um remove the human touch and like you know if it was a craft beer it has a little dirt its character it's a mass-produced one has a little dirt it's dirty so we tolerate um different types of imprecision when

we know that we're dealing with like our neighbor versus when we're expecting some international shipment of uh food um so scott with a question and then looks like you want to share something else stephen you know it's a it's interesting it's kind of the disgust reaction is invoked there too and that idea of um that rejection and anyway the um that reminded me of that last bit was in uh there's a black box of ai that we talk about now well in way we have a black box of industrialization right the thing was made but we don't know how and it used to be that if you look at this if you look at a shoe you can say i see the stitches and i see there's a cow over there i can see how you get leather and i get it then you look at a nike shoe and you're like i have no idea where that came from how it was made if it was glued what was put together right it's just so in a way we have that black box now me when marx was talking about alienation and industrialization maybe part of our alienation is not knowing our connection to craft one of the things i talk to people about is in europe they have these rights called drop morale they're moral rights it's a copyright right overlay so if i sell you a painting in europe you can't destroy it i still have rights as an artist moral rights in the art is a form of expression i think it has a hegelian and kantian background but let's put that aside and then in the united states if i sell you a painting you do anything you want to it we have a lock and utilitarian notion of these expression and rights and individual rights so it feels like there's indifference here philosophically in terms of connection to expressive content in the drought morale context you're not talking about industrial production you're talking about copyright which gets towards industrialization of guild type production right because it's a work of original authorship again maybe not directly related but i just thought it was really interesting that in a way we're looking at that black box question again and humans being alienated from production and i wonder if there's some aspects in in the work that we're seeing here of de-alienating production i wouldn't comment on that nice question scott thank you can you know can you see a diagram it's sort of got a slopey line and some black boxes all right it says here increasing geographical and social distribution of production d growth now d growth centralized industrial production this is happening with premature deindustrialization in in many countries in the world they had some industrialization and now it's it's going and we see that what's coming on is there's a distributed industrial distributed artisanal and distributed diy that's growing what but probably what's still going to come from centralized industrial production is materials form materials and then components local goods is becoming more distributed then distributed infrastructure and then down here this so industrial production so this requires massive capital investment in fixed industrial manufacturing assets then you get

industrial mass production and this is all based upon that you get higher productivity from that massive capital investment but can this produce higher wages more goods purchased is there still scope for will this lead to increased a production employment evidence suggests not so this is non-industrial production for one of a better term so with this there's minimal capital investment in flexible non-industrial production assets more industrial production then there's higher productivity and that's coming and all the options here is you can have higher value artisanal production so if you you know if if shoes are handmade to some extent as they are the company doing that those in ethiopia that can lead to higher wages and that's more goods and infrastructure made with personal commitment leading to increased paid and personal production employment and then down lower price diy production lower expenditure so with presumption there are you could for do produce something with some degree of craft in it and that could produce a higher sales price higher wages and then at the same time your costs can be reduced so if you fabricate your own solar panels and you install them yourselves and suppose you can get all your electricity from that well then your energy costs go down so i'm seeing this has been um is that diagram very visible in your yep i made it large it's what you see on jitsi is not what the live stream is all right so so that diagram i think is addressing both sides of this that on the one hand you're reducing the this lower price diy so there the idea is to eliminate all redundant variation to drive down costs of essential goods as low as possible so as many people can access them as possible so they can um fabricate and install themselves the higher value artisanal production this this is this using uh automated engineering this mass imagineering type of thing pretty so cool reminds me of doing more with less and also bucky um so maybe we can just each have a final thought or a final area that we want to explore a little bit next week and then otherwise this has been an you just think for a few seconds for what would what would be our what was something that will stick with us from today or where do we want to take it in our dot two so um scott first then dean i really like that concluding thought on that really feels like that conversion of labor and attention and and the into embodied our embodied world and that feels like that dynamic is going to be very interesting to continue to explore in that last diagram there those pathways and the complementary pathways are fascinating and thank you for fantastic discussion thank you scott dean yeah i just wanted to i have to read this because i was listening and thinking about what i wanted to close with um my takeaway today besides the fact that i've got a new reference with embedded jet boots that i'm gonna have to go and and try to hold myself down with um is that perhaps life is an apprenticeship uh in wayfinding across scales um so one of

the things that i think that we're trying to do in these live streams popularize active inference we're trying to drive down the costs per se of that entry barrier these kind of interaction processes like we're having today in this point one i think are giving us more of a sense of how that cost factor can be addressed beyond a room full of academics which i think we all agree is kind of a small niche and i think that's something that we're all trying to make available to more people so that last reference that you made scott was actually pretty pretty helpful um i think one of the things that we try to do in terms of an interactionist uh forward approach is in the point zeros we deconstruct and the point ones we reconstruct and then in the point twos which is next tuesday which i'm unfortunately going to be on a i'm going to be away from but that's where the new or next superordinates sort of emerge and so that i think the way that we're trying to set this up whether it's going to i don't know when when it's going to take off like when the jet boots will really be fired up but i think we're putting things in place for that to happen so i think on the one hand i can i can kind of laugh about the fact that i'm going to have to anchor myself but on the other half vladimir says anytime we can get creative with this stuff it's it's worthy it's it's enthusiasm worthy so thank you for sharing with us today i really appreciate it thank you dean um stephen and then blue thank you again yeah i really enjoyed it today is a great great call and um i'm going to take away the idea of situated entropy i think that's really useful and has a bit more a lot of legs to it and i'm really interested in more about talking about pragmatic approaches to using the variation expected free energy i think that goes beyond what was often maybe pragmatism which we've had here as a kind of philosophical thing but maybe pragmatics in terms of design and application so thank you brilliant more than just my closing thought is uh i'm really fascinated by your experience and theorizing and other publications on the past and present and future of that continuum or multi-dimensional space between craft and artisanship the boutique the bespoke everything on that principal component side and then industrialization and have we hit peak something you know have we seen the biggest solid state drives of all time in 2019 that was peak solid state or what will be peak what and what's been going to be what are we at the bottom of what are we near the top of and some of the the roads and the paths that you sketched are pretty cool so steven fox if you have any final comments uh scott could you email me a link to the organization that you you were mentioning i will uh email um i'll email steven's email address to okay thank you very much uh yeah well thank you thank you all very much for taking the trouble to read my paper because it i was plowing a lonely photo when i was doing it lou with the last word thanks steven and you know you gave us so many additional references to read

that that's really what i'm looking forward to doing between now and next week is digging into some of this additional literature that i've been made aware of cool so till next week we'll read the papers think of questions invite any colleagues or anyone else so all backgrounds are welcome to join so we'll see you all next week bye thanks bye bye